NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Balkumari, Lalitpur

Affiliated to Pokhara University

LAB REPORT FOR EMBEDDED SYSTEMS

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LAB REPORT 4: VHDL CODE FOR ADDER

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S. N	LAB TOPIC	DATE	REMARKS
1.	VHDL Code (OR & AND Gate)	20th Nov	
2.	VHDL Code (OR & NOT Gate) using testbench	23rd Nov	
3.	Universal Gates and Special Gates	30th Nov	
4.	VHDL Code for Adder	7th Dec	

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LAB 4: VHDL CODE FOR ADDER

OBJECTIVE:

The objective of this lab is to understand the basic concept of binary addition by designing and implementing **Half Adder** and **Full Adder** circuits using **VHDL**. This lab aims to familiarize students with combinational logic design, Boolean expressions of adders, and VHDL coding for digital arithmetic circuits. It also helps in simulating and verifying the correctness of adder outputs using different input combinations.

THEORY:

Introduction to Adder

An adder is a **combinational digital circuit** that performs the addition of binary numbers. Adders are fundamental components used in arithmetic logic units (ALUs), processors, calculators, and digital systems. Based on the number of input bits, adders are mainly classified into **Half Adder** and **Full Adder**.

Half Adder

A **Half Adder** is a digital circuit that adds **two single-bit binary inputs** and produces two outputs:

- **Sum (S)**
- **Carry (C)**

It does not consider any carry from a previous stage, hence the name *half* adder.

Inputs

- A
- B

Outputs

- Sum (S)
- Carry (C)

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Logic Equations

- **Sum (S) = A ⊕ B**
- **Carry (C) = A · B**

Truth Table

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Half adders are simple but cannot be used alone for multi-bit addition because they do not handle carry-in from previous bits.

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Full Adder

A **Full Adder** is an advanced combinational circuit that adds **three single-bit inputs**:

- Two significant bits
- One carry input from the previous stage (**Cin**)

It produces two outputs:

- **Sum (S)**
- **Carry Out (Cout)**

Inputs

- A
- B
- Cin

Outputs

- Sum (S)
- Carry Out (Cout)

Logic Equations

- **Sum (S) = A ⊕ B ⊕ Cin**
- **Carry (Cout) = (A · B) + (B · Cin) + (A · Cin)**

Truth Table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

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A full adder can be implemented using **two half adders and an OR gate**, making it suitable for cascading to form multi-bit adders.

Adder Implementation Using VHDL

VHDL (VHSIC Hardware Description Language) is used to describe the behavior and structure of digital circuits. In this lab, VHDL is used to model half adder and full adder circuits using **concurrent signal assignments** and **Boolean logic expressions**. Simulation of the VHDL code helps in verifying the correct operation of adders for all possible input combinations.

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VHDL CODE FOR HALF ADDER:

Using Data Flow:

```
library IEEE;
use IEEE.std_logic_1164.all;
Entity ArpHalfAdder is
port(
 A, B : in Std_logic;
 sum, carry : out Std_logic
);
end ArpHalfAdder;
architecture archHalfAdder of ArpHalfAdder is
begin
  sum <= A xor B;
  carry <= A and B;
end archHalfAdder;
```

Using Behavioral:

```
library IEEE;
use IEEE.std_logic_1164.all;
Entity ArpHalfAdder is
port(
 A, B : in Std_logic;
 sum, carry : out Std_logic
);
end ArpHalfAdder;
architecture archHalfAdder of ArpHalfAdder is
begin
  process(A,B)
    begin
  if (A = '0' and B ='0') then
    sum <= '0';
    carry <= '0';
elsif ((A = '1' and B ='0') or (A = '0' and B ='1')) then
    sum <= '1';
    carry <= '0';
  else
    sum <= '0';
    carry <= '1';
  end if;
  end process;
end archHalfAdder;
```

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TEST BENCH FOR HALF ADDER:

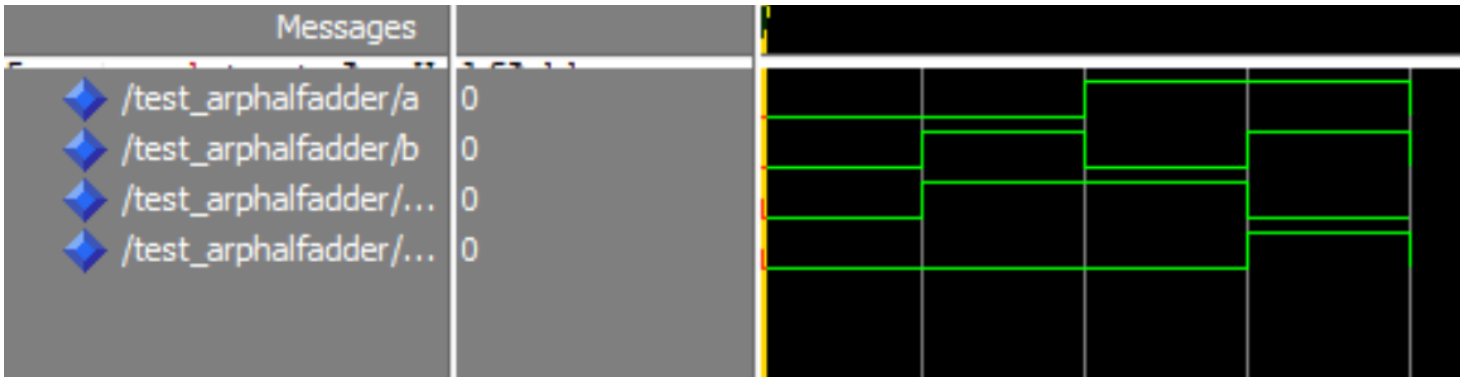
```
library IEEE;
use IEEE.std_logic_1164.all;
entity test_ArpHalfAdder is
end test_ArpHalfAdder;
architecture archTest of test_ArpHalfAdder is
component ArpHalfAdder
  port(
     A, B : in Std_logic;
 sum, carry : out Std_logic
  );
end component;
Signal A,B : std_logic := '0';
Signal sum, carry : std_logic;
begin
  UUT : ArpHalfAdder port map(
  A => A,
  B => B,
  sum => sum,
  carry => carry
  );

  stim_proc : process
  begin
  A <= '0'; B<= '0';
  wait for 100 ps;
  A <= '0'; B<= '1';
  wait for 100 ps;
  A <= '1'; B<= '0';
  wait for 100 ps;
  A <= '1'; B<= '1';
  wait for 100 ps;
end process;
end archTest;
```

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Waveform for Half Adder:



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VHDL CODE FOR FULL ADDER:

USING DATA FLOW:

```
library IEEE;
use IEEE.std_logic_1164.all;

Entity ArpFullAdder is
port(
  A, B, Cin : in Std_logic;
  sum, carry : out Std_logic
);
end ArpFullAdder;

architecture archFullAdder of ArpFullAdder is
begin
  sum <= A xor B xor Cin;
  carry <= (A and B) or (B and Cin) or (A and Cin);
end archFullAdder;
```

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USING BEHAVIORAL:

```
library IEEE;
use IEEE.std_logic_1164.all;

Entity ArpFullAdder is
port(
  A, B, Cin : in Std_logic;
  sum, carry : out Std_logic
);
end ArpFullAdder;

architecture archFullAdder of ArpFullAdder is
begin
  process(A, B, Cin)
  begin
    if (A='0' and B='0' and Cin='0') then
      sum <= '0';
      carry <= '0';
    elsif ((A='0' and B='0' and Cin='1') or
          (A='0' and B='1' and Cin='0') or
          (A='1' and B='0' and Cin='0')) then
      sum <= '1';
      carry <= '0';
    elsif ((A='0' and B='1' and Cin='1') or
          (A='1' and B='0' and Cin='1') or
          (A='1' and B='1' and Cin='0')) then
      sum <= '0';
      carry <= '1';
    else
      sum <= '1';
      carry <= '1';
    end if;
  end process;
end archFullAdder;
```

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TEST BENCH FOR FULL ADDER

```
library IEEE;
use IEEE.std_logic_1164.all;

entity test_ArpFullAdder is
end test_ArpFullAdder;

architecture archTest of test_ArpFullAdder is
component ArpFullAdder
  port(
    A, B, Cin : in Std_logic;
    sum, carry : out Std_logic
  );
end component;

Signal A, B, Cin : std_logic := '0';
Signal sum, carry : std_logic;

begin

  UUT : ArpFullAdder port map(
    A => A,
    B => B,
    Cin => Cin,
    sum => sum,
    carry => carry
  );

  stim_proc : process
  begin
    A <= '0'; B <= '0'; Cin <= '0';
    wait for 100 ps;
    A <= '0'; B <= '0'; Cin <= '1';
    wait for 100 ps;
    A <= '0'; B <= '1'; Cin <= '0';
    wait for 100 ps;
    A <= '0'; B <= '1'; Cin <= '1';
    wait for 100 ps;
    A <= '1'; B <= '0'; Cin <= '0';
    wait for 100 ps;
    A <= '1'; B <= '0'; Cin <= '1';
    wait for 100 ps;
    A <= '1'; B <= '1'; Cin <= '0';
    wait for 100 ps;
    A <= '1'; B <= '1'; Cin <= '1';
    wait for 100 ps;
  end process;

end archTest;
```

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RESULT:

The Half Adder and Full Adder circuits were successfully designed and implemented using **VHDL**. Both **data flow** and **behavioral** modeling techniques were used, and the designs were verified using appropriate **test benches**. The simulation waveforms showed correct **Sum** and **Carry** outputs for all possible input combinations, matching the respective truth tables of Half Adder and Full Adder.

CONCLUSION:

This practical helped in understanding the working of **binary adders** and their implementation using **VHDL**. Through this experiment, the difference between **Half Adder and Full Adder** was clearly observed, especially the role of carry input in the Full Adder. The use of data flow and behavioral models improved understanding of different VHDL coding styles. The successful simulation confirms that VHDL can be effectively used to design and verify basic arithmetic circuits used in digital systems.

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